

Maternal Health Risk Stratification

Statistical Analysis, Exploratory Data Analysis, and Interpretable
Machine Learning

Community StudyBuild

Health & Medical Data Science Project

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Abstract

This article presents a reproducible statistical and machine-learning analysis of the UCI Maternal Health Risk dataset, which contains physiological measurements and a categorical maternal risk label collected through an IoT-based monitoring system in rural Bangladesh. The analytical objective is to characterize differences among Low, Mid, and High Risk groups and evaluate whether a simple, interpretable classifier can identify patients in the High Risk category. The final analytical sample contains 1,012 observations after removal of two physiologically implausible heart-rate records. Exact duplicate rows were retained because coarse sensor rounding can make distinct observations identical; treating them as duplicate patients would risk distorting the class distribution.

The analysis combines descriptive statistics, grouped summaries, boxplots, scatter plots, threshold-based checks, SQL-based reproducibility, and two baseline classifiers: multinomial Logistic Regression and a shallow Decision Tree. Blood Sugar and blood pressure show the strongest empirical separation among risk groups, whereas Heart Rate provides comparatively little separation. On an 80/20 stratified hold-out test set, the Decision Tree achieved 0.680 accuracy, 0.670 macro F1, 0.836 High-Risk Recall, and 0.920 High-Risk Precision, outperforming Logistic Regression on the reported metrics. The principal modeling challenge was Low-versus-Mid risk discrimination, while the selected tree captured High Risk relatively well.

The results demonstrate how transparent statistical analysis can complement machine learning in a health-data setting. However, the study is observational, geographically narrow, based on a small set of physiological variables, and potentially affected by target-label construction. Therefore, the model should be interpreted as an educational screening baseline, not as a validated clinical decision system.

Keywords: maternal health; risk stratification; descriptive statistics; blood sugar; blood pressure; logistic regression; decision tree; classification; interpretable machine learning; Python; SQL.

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1 Introduction

Maternal health remains an important public-health and data-science problem because risk can change rapidly during pregnancy and because timely identification of high-risk patients may support earlier clinical assessment. Digital health systems and low-cost physiological sensors create opportunities to analyze measurements such as blood pressure, blood glucose, body temperature, and heart rate at scale. The UCI Maternal Health Risk dataset was collected from hospitals, community clinics, and maternal-health settings in rural Bangladesh through an IoT-based risk-monitoring system [1, 2].

The present study is designed as an applied statistical learning exercise rather than a clinical validation study. The central question is:

How strongly do a small set of routinely recorded physiological variables distinguish Low, Mid, and High maternal risk, and can an interpretable baseline classifier identify High-Risk observations with useful sensitivity?

The project follows a deliberately transparent workflow. First, the raw data are audited for missingness, duplicates, and implausible physiological values. Second, descriptive statistics and visualizations are used to compare risk groups. Third, simple classification models are trained using a fixed, stratified split. Finally, model performance is interpreted using class-specific precision, recall, F1 score, and confusion matrices rather than accuracy alone.

The emphasis on interpretability is particularly important in health analytics. A model can produce a high aggregate score while failing an important subgroup. In a risk-stratification context, missing a genuinely High-Risk observation may be more consequential than confusing two lower-risk categories. Accordingly, High-Risk Recall is treated as a primary evaluation criterion, with macro F1 and other metrics providing complementary information.

2 Data and Study Design

2.1 Data source and variables

The dataset used in this project is the UCI Maternal Health Risk dataset (Dataset 863), which contains 1,014 observations and six predictor/target fields. The UCI documentation reports that the data were collected in rural Bangladesh using an IoT-based risk monitoring system and that the dataset has no missing values [1]. The official UCI citation identifies the dataset as Ahmed (2020), DOI 10.24432/C5DP5D [1].

Table 1: Variables used in the analysis.

Variable	Type / Unit	Description
Age	Numeric; years	Maternal age recorded in years.
SystolicBP	Numeric; mmHg	Systolic blood pressure.
DiastolicBP	Numeric; mmHg	Diastolic blood pressure.
BS	Numeric; mmol/L	Blood sugar concentration.
BodyTemp	Numeric; °F	Body temperature.
HeartRate	Numeric; bpm	Heart rate.
RiskLevel	Categorical	Low Risk, Mid Risk, or High Risk.

The target variable is categorical with three levels: *low risk*, *mid risk*, and *high risk*. The analysis therefore treats the task as multiclass classification. Importantly, the variables are not independent clinical

determinants of maternal outcomes: they are measurements associated with a pre-existing risk label. The interpretation is therefore associative rather than causal.

2.2 Analytical sample and data cleaning

The source data contain no missing values, so imputation was unnecessary. An audit found 562 exact duplicate rows among the cleaned input records, representing approximately 55.4% of the observations. These rows were deliberately retained. The source variables are rounded coarsely, particularly blood pressure, blood sugar, and body temperature, meaning that different patients can legitimately share the same recorded values. Removing all exact duplicates would therefore assume that identical measurements represent the same patient and could artificially change the class balance.

Two observations had a HeartRate value of 7 bpm. These values were considered physiologically implausible for the monitored population and were removed as likely sensor or data-entry errors. No artificial replacement value was imputed because the true measurement is unknown. The final analytical sample consequently contained 1,012 observations. The raw project CSV contains 1,014 observations; exactly two records have HeartRate = 7 bpm and are removed. Therefore, every downstream table, statistical test, model, and dashboard in this article uses the same 1,012-row analytical dataset (Low Risk = 404, Mid Risk = 336, High Risk = 272).

Table 2: Data-quality decisions.

Check	Finding	Decision
Missing values	0	No imputation.
Exact duplicates	562	Retained because coarse rounding can create legitimate identical rows.
Implausible heart rate	2 rows with HR = 7 bpm	Removed as likely sensor/entry errors.
Other unusual ages/values	Present	Reported as a limitation rather than silently corrected.
Final sample	1,012	Used for all downstream analyses.

3 Exploratory Data Analysis

3.1 Risk-level distribution

After cleaning, the sample consisted of 404 Low-Risk, 336 Mid-Risk, and 272 High-Risk observations. The corresponding percentages were 39.9%, 33.2%, and 26.9%. This is not an extreme class imbalance, but the classes are sufficiently different in size that model evaluation should not rely on accuracy alone.

Table 3 shows the group-level means. High-Risk observations have higher mean age, systolic blood pressure, diastolic blood pressure, blood sugar, body temperature, and heart rate than Low-Risk observations. Among these variables, the difference in blood sugar is particularly pronounced.

Table 3: Group-level descriptive statistics.

Risk	n	Age	SBP	DBP	BS	HR
Low	404	26.9	105.8	72.5	7.22	73.1
Mid	336	28.4	113.2	74.2	7.80	74.2
High	272	36.2	124.2	85.1	12.12	76.7

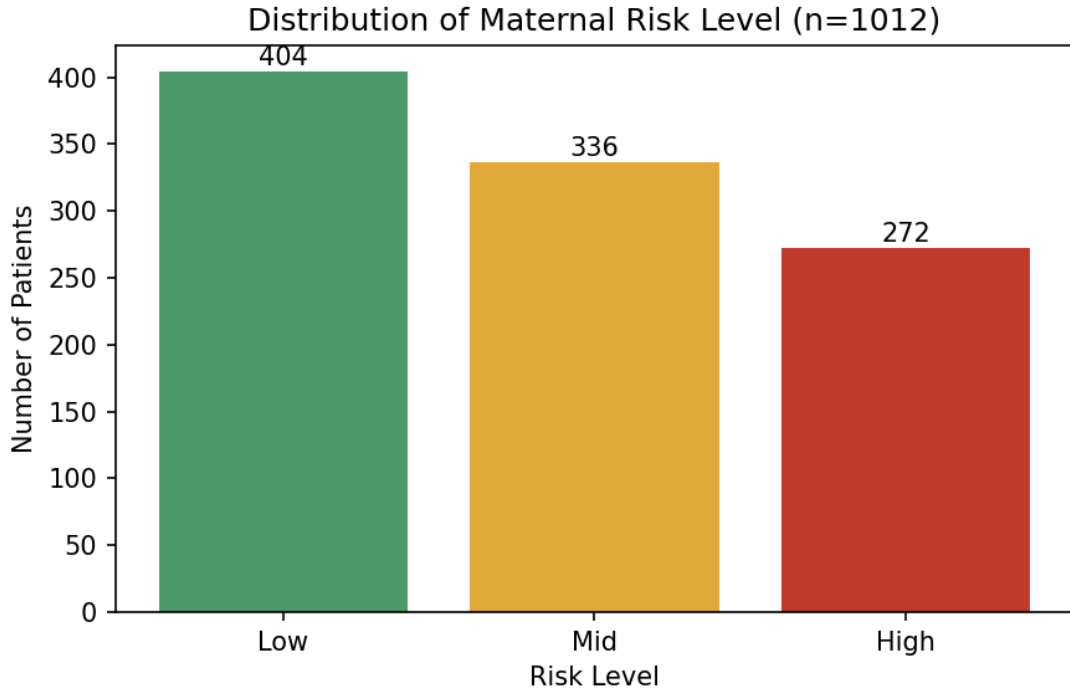


Figure 1: Distribution of Low, Mid, and High maternal risk observations.

3.2 Descriptive statistical framework

For a variable X with observations $x_1, \dots, x_n$, the sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i. \quad (1)$$

The sample standard deviation is

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}. \quad (2)$$

The interquartile range (IQR) is

$$IQR = Q_3 - Q_1, \quad (3)$$

where Q_1 and Q_3 are the 25th and 75th percentiles. Boxplots were used because they display the median, central spread, and unusually extreme observations without requiring strong distributional assumptions.

3.3 Blood sugar

Blood Sugar (BS) is the clearest single variable separating High-Risk observations from the other two classes. The High-Risk group has a mean of 12.12 mmol/L compared with 7.22 mmol/L for Low Risk and 7.80 mmol/L for Mid Risk. The standard deviations are also substantially different: 0.64, 2.28, and 4.17 for Low, Mid, and High Risk, respectively.

A threshold-based analysis further illustrates this separation. Among High-Risk observations, 170 of 272 (62.5%) met the project's high-blood-sugar threshold, compared with only 4 of 404 (1.0%) Low-Risk observations and 32 of 336 (9.5%) Mid-Risk observations. This does not prove that high blood sugar causes

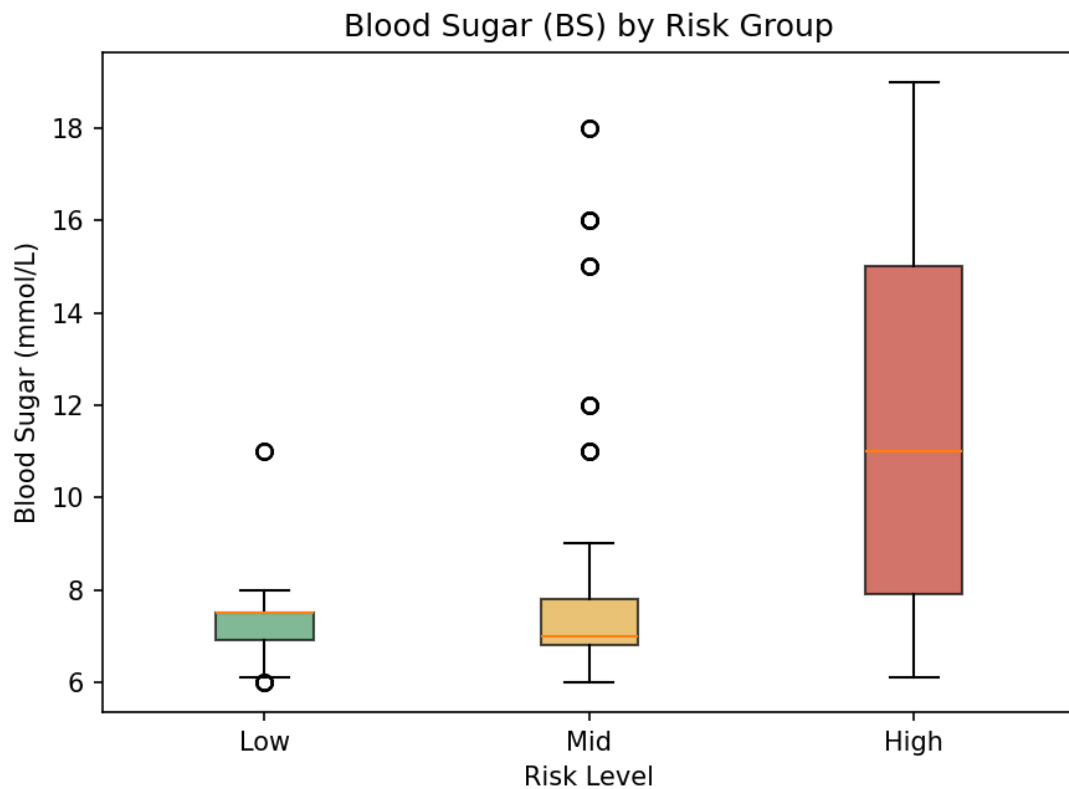


Figure 2: Blood Sugar distribution by maternal risk group.

the assigned risk label; rather, it shows that the two variables are strongly aligned in this dataset.

3.4 Blood pressure

Both systolic and diastolic blood pressure increase across risk groups. Mean systolic blood pressure rises from 105.8 mmHg in Low Risk to 124.2 mmHg in High Risk, while mean diastolic blood pressure rises from 72.5 to 85.1 mmHg.

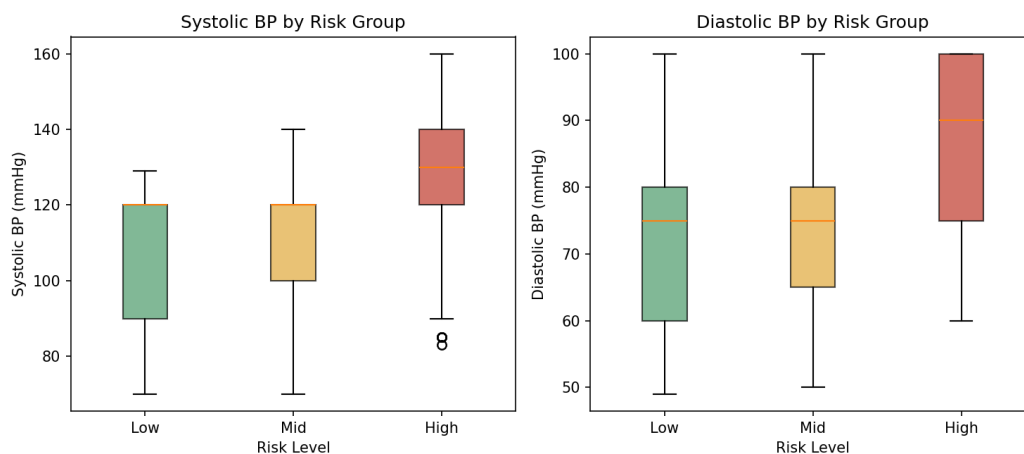


Figure 3: Blood pressure distributions across maternal risk groups.

The group separation is less clean than for Blood Sugar. Low and Mid Risk overlap substantially, and some High-Risk observations occur in the same numerical region as lower risk observations. This overlap is important because it indicates that no single variable provides a perfect deterministic classifier.

3.5 Age and heart rate

Age shows a positive shift toward the High-Risk group: the mean increases from 26.9 years in Low Risk to 36.2 years in High Risk. Nevertheless, the distribution remains heterogeneous, and unusual ages are present in the source data. These observations were retained because there was insufficient information to determine whether they represented genuine cases, measurement issues, or data-entry artifacts.

Heart Rate shows comparatively weak separation. The group means are 73.1, 74.2, and 76.7 bpm. The distributions overlap considerably, suggesting that heart rate contributes less information to this classification task than blood sugar or blood pressure.

3.6 Joint relationship between blood sugar and systolic blood pressure

A two-dimensional scatter plot provides a useful complement to univariate summaries. High-Risk observations become more concentrated toward higher values of both Blood Sugar and SystolicBP, but there is no perfect boundary between classes.

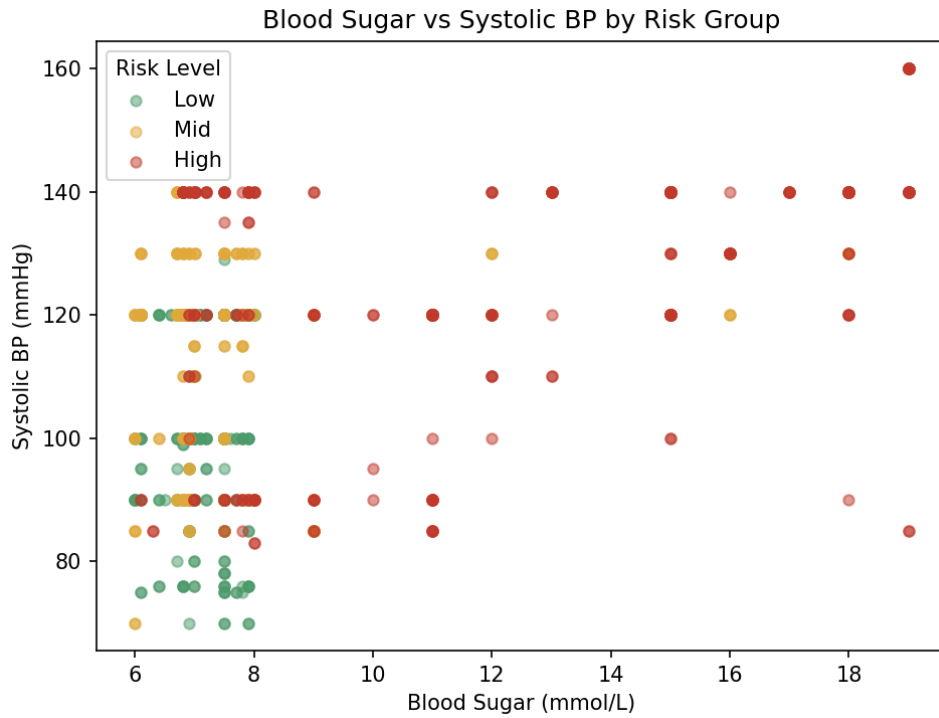


Figure 4: Joint relationship between Blood Sugar and Systolic Blood Pressure.

This pattern motivates a central methodological caution: the data support statistical association, not causal inference. In addition, the original RiskLevel label may itself have been influenced by blood pressure and blood sugar thresholds. Therefore, a model that performs well using these variables may partly be reproducing the structure of the labeling process rather than independently discovering new clinical risk mechanisms.

4 Statistical and Machine-Learning Methods

4.1 Reproducible analytical pipeline

The project was implemented using Python and supplemented by SQLite SQL queries. The main workflow was:

1. Load the UTF-8-signed CSV and normalize categorical labels.
2. Audit missingness, duplicates, and implausible measurements.
3. Remove only the two implausible HeartRate records.
4. Compute descriptive statistics by RiskLevel.
5. Produce class-distribution, boxplot, and scatter visualizations.
6. Create an 80/20 stratified train/test split using `random.state=42`.
7. Fit StandardScaler using training data only.
8. Train multinomial Logistic Regression and a shallow Decision Tree.
9. Evaluate class-specific performance using confusion matrices and precision, recall, and F1 scores.
10. Select the final baseline using High-Risk Recall first and macro F1 as a secondary criterion.

4.2 Stratified train/test split

The cleaned data were divided into training and test subsets using an 80/20 stratified split. Stratification preserves approximately the same Low/Mid/High proportions in each subset. The test set therefore contains 203 observations, including 55 High-Risk cases.

For a binary class, recall is defined as

$$Recall = \frac{TP}{TP + FN}, \quad (4)$$

where TP denotes true positives and FN denotes false negatives. For High Risk, the interpretation is particularly direct: it estimates the proportion of truly High-Risk cases that the model successfully flags.

Precision is

$$Precision = \frac{TP}{TP + FP}, \quad (5)$$

and the F1 score is the harmonic mean

$$F_1 = 2 \frac{Precision \cdot Recall}{Precision + Recall}. \quad (6)$$

For multiclass classification, macro F1 is calculated as the unweighted mean of the class specific F1 scores:

$$F_{1,macro} = \frac{1}{K} \sum_{k=1}^K F_{1,k}, \quad (7)$$

where $K = 3$ for Low, Mid, and High Risk.

Accuracy remains useful as a broad descriptive metric,

$$Accuracy = \frac{\sum_{k=1}^K TP_k}{N}, \quad (8)$$

but it is not sufficient because it can hide poor performance in a particular class.

4.3 Logistic Regression

Multinomial Logistic Regression provides a transparent linear baseline. For class k , the model estimates a score

$$z_k = \beta_{0k} + \beta_{1k}x_1 + \cdots + \beta_{pk}x_p, \quad (9)$$

and converts class scores into probabilities through the softmax function

$$P(Y = k | X) = \frac{\exp(z_k)}{\sum_{j=1}^K \exp(z_j)}. \quad (10)$$

Standardization was applied for Logistic Regression because the predictors are measured in different units. Crucially, the scaler was fitted only on the training data and then applied to the test data. This prevents information from the test set from leaking into model preparation.

4.4 Decision Tree

The second model was a Decision Tree with maximum depth 4 and minimum leaf size 15. The tree was intentionally kept shallow to preserve interpretability and reduce overfitting risk in a small dataset.

At each candidate split, a classification tree can use the Gini impurity

$$Gini(t) = 1 - \sum_{k=1}^K p_k^2, \quad (11)$$

where p_k is the proportion of class k within node t . A good split reduces impurity in the resulting child nodes.

The final tree’s feature importance values are model-based measures of how much each variable contributes to impurity reduction across the tree. They should not be interpreted as causal effects.

5 Machine-Learning Results

5.1 Model comparison

The two baseline models produced noticeably different results. Logistic Regression achieved 0.586 accuracy, 0.584 macro F1, 0.782 High-Risk Recall, and 0.768 High-Risk Precision. The Decision Tree achieved 0.680 accuracy, 0.670 macro F1, 0.836 High-Risk Recall, and 0.920 High-Risk Precision.

Table 4: Performance comparison on the held-out test set.

Model	Accuracy	Macro F1	High Recall	High Precision
Logistic Regression	0.586	0.584	0.782	0.768
Decision Tree	0.680	0.670	0.836	0.920

The Decision Tree was selected because it outperformed Logistic Regression on all four reported metrics. Most importantly, the High-Risk Recall increased from 0.782 to 0.836, while High-Risk Precision increased from 0.768 to 0.920. Thus, among the two simple baselines, the tree provides a stronger balance between catching High-Risk observations and avoiding unnecessary High-Risk flags.

Table 5: Class-wise performance of the selected Decision Tree.

Class	Precision	Recall	F1	Support
Low Risk	0.614	0.864	0.718	81
Mid Risk	0.564	0.328	0.415	67
High Risk	0.920	0.836	0.876	55

5.2 Class-wise Decision Tree performance

The class-wise results reveal a more nuanced picture than the overall accuracy. The model is strongest for High Risk and reasonably strong for Low Risk, but Mid Risk is substantially harder to identify. This is an important result: the primary modeling weakness is not failure to recognize all high-risk patients, but the substantial overlap between Low and Mid Risk.

5.3 Confusion matrices

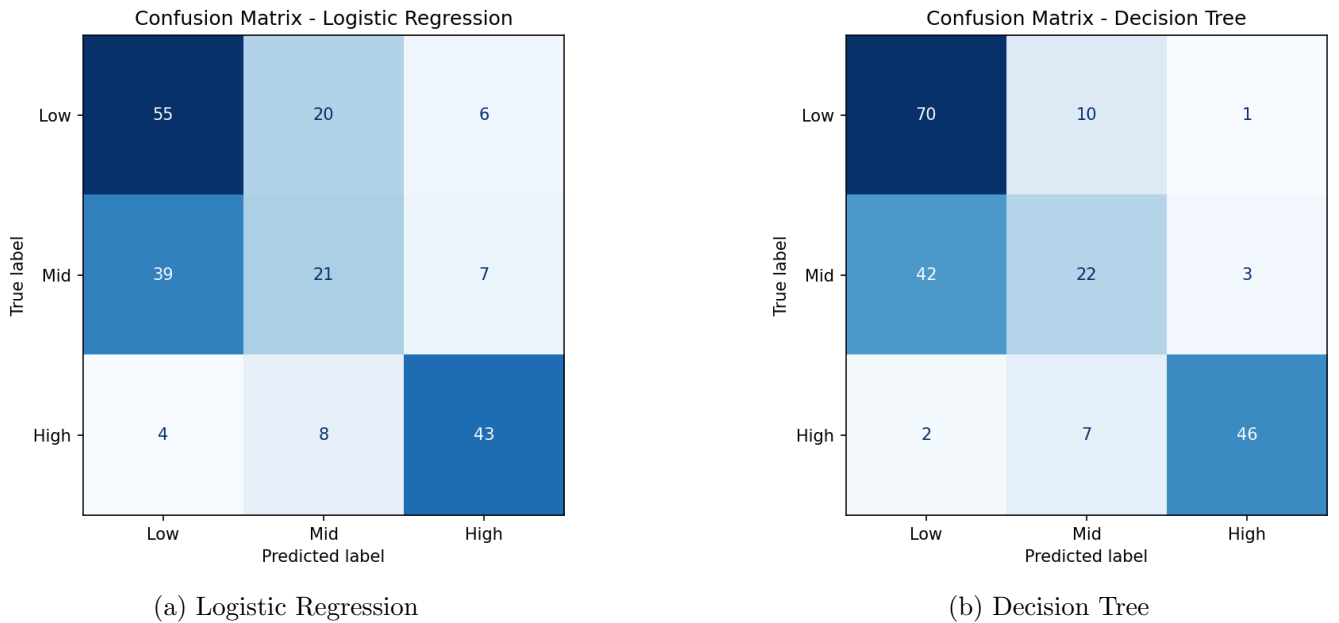


Figure 5: Confusion matrices for the two baseline classifiers.

The error pattern is clinically relevant. The Decision Tree misclassified 32% of the test cases. Among the errors, 42 of 65 were Mid-Risk observations predicted as Low Risk. In contrast, 9 of 55 true High-Risk cases were predicted as a lower-risk category. Thus, the model still misses some High-Risk patients even though its recall is relatively high.

The High-Risk recall of 0.836 can be interpreted as follows: approximately 83.6% of the true High-Risk test observations were correctly flagged. Equivalently, approximately 16.4% were not captured by the model in this particular test split. For a clinical application, that remaining false-negative rate would be too important to ignore.

6 Feature Interpretation

6.1 Feature importance

The selected Decision Tree assigns approximately 50% of its feature importance to Blood Sugar and approximately 33% to Systolic Blood Pressure. The remaining variables have much smaller contributions.

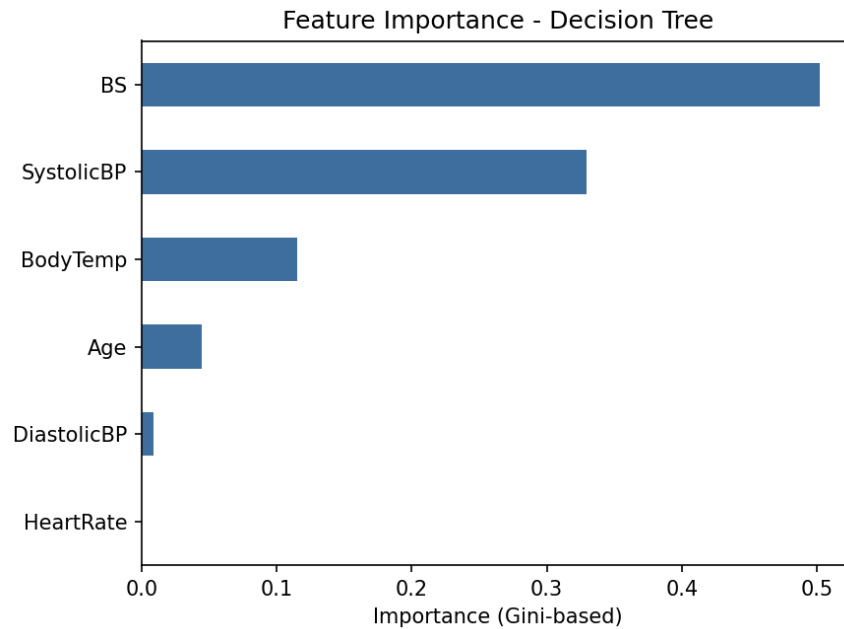


Figure 6: Decision Tree feature importance.

The ranking is consistent with the descriptive EDA. Blood Sugar produces the strongest univariate separation, while systolic blood pressure provides a secondary but substantial signal. Diastolic blood pressure, Age, and Heart Rate contribute less to the tree’s splitting decisions.

A feature-importance score should not be interpreted as an effect size, odds ratio, or causal coefficient. It indicates how much the fitted tree used a variable to reduce impurity. The importance can therefore be affected by the target-label construction, measurement scale, correlations among predictors, and the particular train/test sample.

6.2 Decision rules

The explicit tree structure is valuable because it makes the model inspectable. Instead of presenting a black-box probability, each prediction can be traced through a sequence of threshold-based decisions. This is especially appropriate for an educational health-data project where transparent reasoning is preferred over marginal performance gains from complex ensemble methods.

7 SQL Analysis and Reproducibility

A complementary SQLite layer was included so that core exploratory findings can be reproduced without relying exclusively on pandas. The cleaned 1,012-row dataset is stored in a single SQLite table, and nine documented queries reproduce class distributions, summary statistics, group comparisons, threshold checks, and age-band cross-tabulations.

The SQL implementation uses standard relational operations including `GROUP BY`, `CASE WHEN`, common

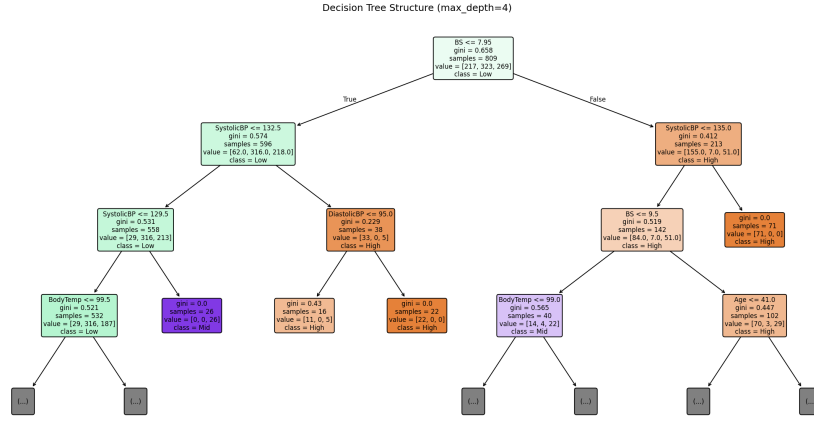


Figure 7: Structure of the shallow Decision Tree used as the final baseline.

table expressions, and window functions. Because SQLite does not provide a native median function in the same way as some analytical database systems, the project uses `ROW_NUMBER()` as part of its ordered group logic. Population standard deviation can also be reconstructed from

$$\sigma = \sqrt{E(X^2) - [E(X)]^2}. \quad (12)$$

The SQL layer is particularly useful for demonstrating that the major descriptive findings are not dependent on a single software interface. It also makes the project easier to inspect for users who work primarily with databases.

Reproducibility was strengthened through three design choices. First, the random seed was fixed at 42 wherever a stochastic process was used. Second, both a Python script and a notebook implement the same analytical pipeline. Third, the project includes explicit package requirements and documented preprocessing decisions.

8 Analytical Dashboard

To communicate the statistical findings more effectively, the project includes an advanced analytics dashboard rather than relying only on the simpler overview presentation. The advanced dashboard brings together population structure, clinical distributions, risk intensity, model interpretation, and model-performance indicators in one visual analytical workspace. This design is particularly useful because the project is not limited to a single descriptive question: it asks which physiological variables separate risk groups, where the highest-risk observations are concentrated, and how well an interpretable classifier performs.

8.1 Dashboard components and analytical value

The first row of the dashboard provides decision-oriented summary cards. The “Primary Drivers” card highlights Blood Sugar and Systolic Blood Pressure as the dominant model drivers, while the “Hardest Call” card emphasizes the substantial overlap between Low and Mid Risk. The “Safety Gap” card communicates that false negatives remain important even though the classifier identifies a large proportion of High-Risk observations. Finally, the “Model Status” card explicitly frames the Decision Tree as a screening aid rather than a clinical decision system.

The middle section expands the descriptive analysis. The risk-volume lollipop shows that Low Risk is the largest group (404 observations), followed by Mid Risk (336) and High Risk (272). The violin plot



Maternal Health Risk — Advanced BI Visuals

Added visual for Section 8 — Analytical Dashboard

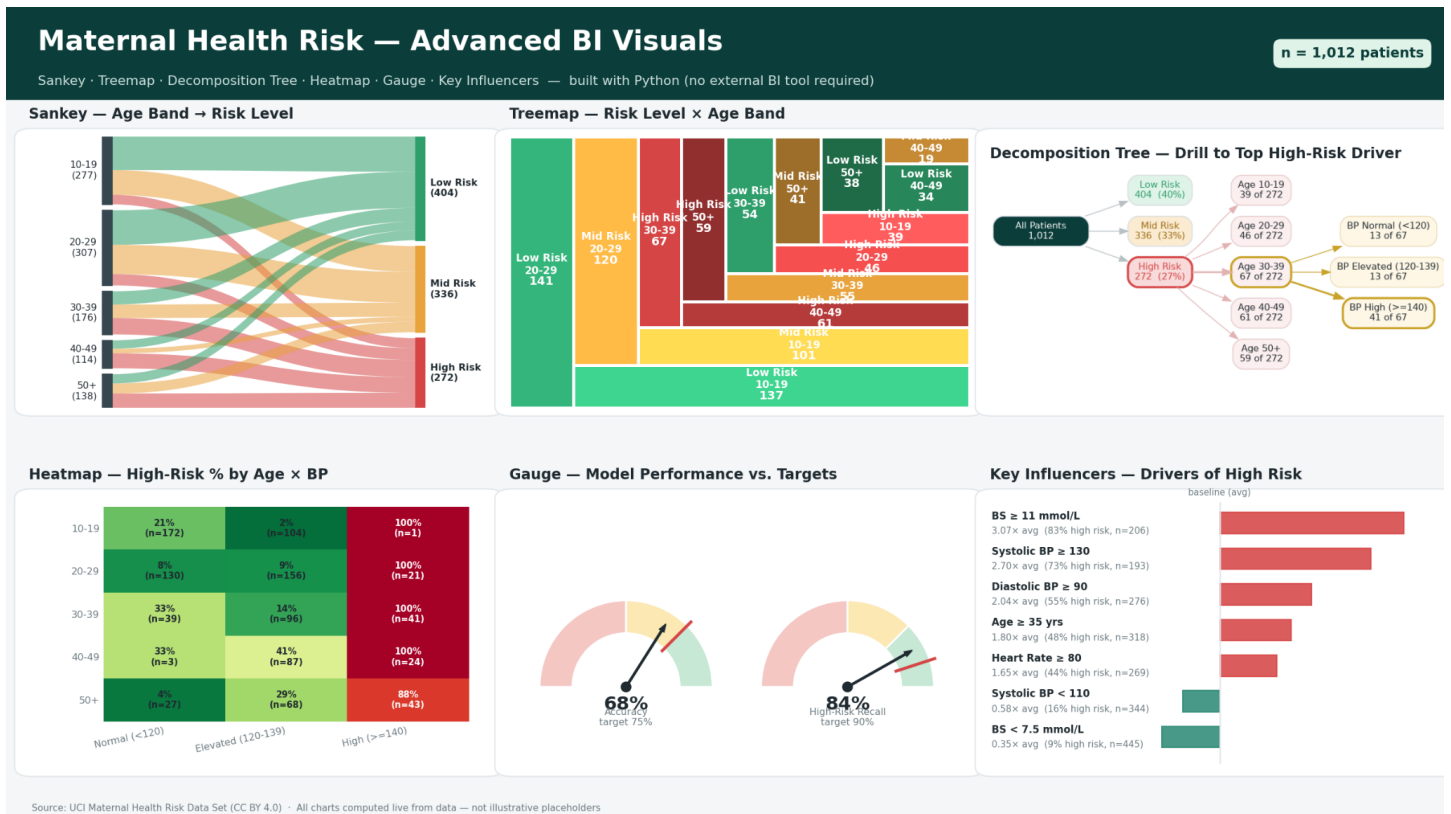


Figure 8A. Advanced BI dashboard combining Sankey, Treemap, Decomposition Tree, Heatmap, Gauge, and Key Influencers views. The dashboard uses the same 1,012-observation analytical sample described in the article. It is an analytical/educational visualization and not a clinical decision-making tool.

and ECDF provide complementary views of age: the violin plot shows the shape and concentration of observations within each risk category, while the ECDF makes between-group differences in cumulative age distributions easier to compare. Together, these views show that age alone cannot explain the observed risk stratification.

The area chart adds a particularly useful population-level perspective by displaying the High-Risk share across age bands. The reported high-risk proportions increase substantially from the younger groups into the 40–59 year bands before decreasing in the 60–69 group. Because the oldest band contains fewer observations than some middle bands, the apparent decrease should be interpreted cautiously rather than as evidence of a protective effect.

The bubble chart connects systolic blood-pressure bands with High-Risk intensity and patient volume. This is useful because a percentage alone can be misleading when the denominator is small. The dashboard therefore communicates both risk intensity and the number of patients contributing to each band. The radar chart provides a normalized profile comparison across Age, Systolic BP, Diastolic BP, Blood Sugar, Body Temperature, and Heart Rate. Its purpose is comparative pattern recognition, not inference from the polygon area.

8.2 Model interpretation and performance view

The lower analytical section makes the machine-learning component more transparent. The feature-importance chart ranks Blood Sugar at approximately 50.2%, Systolic BP at 33.0%, Body Temperature at 11.5%, Age at 4.5%, Diastolic BP at 0.8%, and Heart Rate at 0.0%. This ordering is consistent with the statistical exploration: Blood Sugar and blood pressure are the variables most strongly associated with the risk categories in the available data.

The bullet scorecard reports approximately 68.0% classification accuracy and 83.6% High-Risk Recall. The target markers shown in the dashboard are illustrative screening-aid benchmarks rather than clinical acceptance thresholds. Consequently, the visual should be read as a performance-monitoring device, not as evidence that the model meets a medical standard.

The bottom range panels compare means and medians for Systolic BP, Diastolic BP, Blood Sugar, and Heart Rate. These panels are valuable for identifying skewness and for checking whether the mean is being pulled away from the median. In particular, the Blood Sugar panel reinforces the central role of glucose-related measurements in differentiating risk categories.

Overall, the advanced dashboard extends the article’s statistical results into a compact decision-support narrative: *what is the population structure, which physiological signals matter most, where is High Risk concentrated, and how reliable is the classifier?* It complements the formal statistical tables and model metrics without replacing them. The underlying sample definition, model-validation procedure, and study limitations therefore remain essential when interpreting every dashboard element.

9 Expanded Results and Integrated Interpretation

The combined statistical and machine-learning results provide a consistent picture of maternal risk stratification in this dataset. The descriptive analysis first establishes that the three risk groups are not simply separated by one universal threshold. Instead, several physiological variables shift together, with the largest differences appearing in Blood Sugar and blood pressure. The machine-learning analysis then evaluates whether these multivariable patterns can be converted into an interpretable classification rule.

9.1 Quantitative summary of the main findings

Table 6 summarizes the most important numerical findings. The table is intentionally compact so that the reader can compare population structure, clinical separation, and predictive performance in one place.

Table 6: Integrated summary of principal analytical results.

Component	Result	Interpretation
Final analytical sample	1,012	Two physiologically implausible heart-rate records removed
Risk distribution	404 / 336 / 272	Low / Mid / High Risk observations
Low-Risk Blood Sugar mean	7.22 mmol/L	Lowest group-level glucose level
High-Risk Blood Sugar mean	12.12 mmol/L	Strong upward shift with risk category
Decision Tree accuracy	0.680	Moderate overall hold-out classification performance
Decision Tree macro F1	0.670	Balanced summary across the three classes
High-Risk Recall	0.836	83.6% of High-Risk test observations detected
High-Risk Precision	0.920	Most observations predicted High Risk were truly High Risk
Top feature: Blood Sugar	50.2%	Largest Decision Tree feature importance
Second feature: Systolic BP	33.0%	Major additional contributor to the learned rules

The difference between High-Risk Recall and Precision is especially important. Recall measures the proportion of true High-Risk observations detected,

$$\text{Recall}_H = \frac{TP_H}{TP_H + FN_H}, \quad (13)$$

where TP_H and FN_H denote true positives and false negatives for the High-Risk class. Precision instead measures the reliability of High-Risk alerts,

$$\text{Precision}_H = \frac{TP_H}{TP_H + FP_H}. \quad (14)$$

The observed values of 0.836 and 0.920 therefore indicate a useful screening pattern in this dataset: the model detects most High-Risk observations and, among its High-Risk predictions, a relatively large proportion are correct. Nevertheless, nine of the 55 High-Risk observations in the test set were still missed. In a clinical context, that residual false-negative count would require explicit cost-sensitive evaluation.

9.2 Why Blood Sugar and blood pressure dominate

The feature-importance results are supported by the descriptive plots. Blood Sugar has both a large between-group shift and substantial contribution to the Decision Tree. Systolic Blood Pressure provides the second-largest contribution. Body Temperature has a smaller but visible role, while Age and Diastolic Blood Pressure contribute relatively little to the fitted tree and Heart Rate contributes essentially none.

This agreement between independent analytical views is important. A variable that appears important only in a model can be an artifact of the fitted algorithm; here, the model ranking is broadly consistent with the grouped distributions and dashboard summaries. The convergence of evidence strengthens the conclusion that glucose and systolic blood pressure are the most informative variables available in this small dataset.

9.3 Classification errors and practical meaning

The confusion-matrix analysis shows that the principal weakness is not High-Risk detection but the separation of Low and Mid Risk. A large share of the observed errors involves Mid-Risk cases being classified as Low Risk. This indicates that the boundary between these categories is diffuse and that a three-class decision may be intrinsically harder than identifying the most extreme risk group.

For a screening-oriented application, this suggests that future work should evaluate alternative decision strategies. For example, a binary “High Risk versus not High Risk” formulation could be assessed separately from the three-class problem. Such an analysis should still preserve calibration, uncertainty, and clinically meaningful costs rather than optimizing accuracy alone.

9.4 Statistical evidence and machine-learning evidence together

The main result is therefore not a single accuracy number. The stronger conclusion is that multiple forms of evidence point in the same direction: risk groups differ in important physiological measurements; Blood Sugar and Systolic BP dominate the interpretable model; the Decision Tree captures High Risk better than Logistic Regression; and the remaining errors are concentrated around the Low–Mid boundary.

This integrated interpretation is more informative than model performance alone because it connects the predictive result to the observed data structure. It also clarifies what future models should improve: better discrimination between adjacent risk groups, robust validation, probability calibration, and independent clinical outcomes.

10 Discussion

11 Discussion

11.1 Principal findings

Three findings are particularly important. First, Blood Sugar is the strongest single statistical marker in this dataset. Its mean rises from 7.22 mmol/L in Low Risk to 12.12 mmol/L in High Risk, and its distribution shows a much larger shift than Heart Rate or Body Temperature. Second, blood pressure variables show a clear but less complete stepwise relationship with risk. Third, Low and Mid Risk overlap substantially, creating a classification problem that cannot be solved reliably by one simple threshold.

The Decision Tree’s performance supports the value of nonlinear, threshold-based rules in this setting. Its 0.680 accuracy is materially higher than Logistic Regression’s 0.586 accuracy, and its High-Risk Recall of 0.836 is particularly relevant to the project’s stated objective. However, the result should not be interpreted as evidence that the tree is clinically ready. The test set is small, the features are limited, and the target label may already encode information from the same measurements.

11.2 Clinical interpretation versus statistical association

The analysis is consistent with the idea that blood glucose and blood pressure are important signals for maternal risk. Nevertheless, statistical association cannot establish causation. The model learns relationships between recorded measurements and the existing RiskLevel label; it does not observe future maternal outcomes such as complications, hospitalization, or mortality.

A particularly important issue is possible label leakage in a conceptual sense. If the original risk categories were assigned partly by rules involving blood pressure or blood sugar, then strong model performance using

these same variables is expected. The model may therefore be reproducing a labeling protocol rather than independently predicting an external clinical outcome.

11.3 Error analysis

The confusion matrix indicates that Mid Risk is the most difficult class. Forty-two of the reported 65 classification errors were Mid Risk cases predicted as Low Risk. This suggests that the boundary between lower-risk categories is less distinct than the High-Risk boundary.

The model still produced nine false-negative High-Risk predictions in the 203-observation test set. A screening model intended for real clinical use would need explicit cost-sensitive evaluation because the consequences of false negatives and false positives are not equivalent.

12 Limitations

Several limitations constrain the generalizability of the findings.

1. **Geographic scope.** The source data come from rural Bangladesh and may not represent other healthcare systems or populations.
2. **Limited predictors.** The dataset contains only six fields, of which five are predictors. It does not include gestational age, obstetric history, laboratory panels, comorbidities, medication use, or socioeconomic variables.
3. **Duplicate measurements.** A large number of exact duplicate rows remain in the analysis because the measurements are coarsely rounded. This reduces the effective diversity of the sample.
4. **Potential label dependence.** Risk labels may have been constructed using some of the same physiological variables supplied to the classifier.
5. **No external validation.** The Decision Tree was evaluated on one held-out test set rather than on an independent cohort.
6. **No prospective clinical endpoint.** The target is RiskLevel rather than a future, independently observed clinical outcome.
7. **Small test set.** Only 55 High-Risk observations appear in the test split, so estimates such as High-Risk Recall are subject to sampling variability.
8. **No clinical deployment assessment.** The project does not evaluate calibration, decision-curve utility, clinical workflow integration, fairness across demographic subgroups, or regulatory requirements.

These limitations are not minor technical details; they define the appropriate scope of the conclusions. The project is best understood as an interpretable analytical baseline.

13 Recommendations and Future Work

A stronger research study should begin by collecting a larger and more diverse cohort with clear prospective clinical outcomes. The data should include gestational age, obstetric history, previous complications, relevant laboratory measurements, and clinically validated outcomes.

Second, future modeling should use repeated stratified cross-validation and report uncertainty intervals for key metrics. A useful extension would be to estimate a confidence interval for High-Risk Recall. For a proportion $\hat{p}$, an approximate Wald interval is

$$\hat{p} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \quad (15)$$

although exact or bootstrap intervals may be preferable for small subgroup counts.

Third, the modeling objective should be explicitly cost-sensitive. If missing High Risk is much more costly than generating a false alarm, model selection should reflect that asymmetric cost. A general expected loss can be written as

$$\mathcal{L} = \sum_{i=1}^N C(y_i, \hat{y}_i), \quad (16)$$

where $C(y_i, \hat{y}_i)$ assigns a higher penalty to clinically consequential errors.

Fourth, future work should compare interpretable trees with carefully validated regularized models, calibrated probabilities, and, if justified by sample size, more advanced algorithms. However, model complexity should only be increased when it provides demonstrable and validated benefit.

Finally, external validation and clinician review are essential before any clinical deployment. A model should never replace professional assessment merely because it achieves a strong score on a historical dataset.

14 Conclusion

This study demonstrates a complete, reproducible workflow for maternal health risk stratification using descriptive statistics, visualization, SQL, and interpretable machine learning. After minimal and transparent cleaning, the final sample contained 1,012 observations distributed across Low, Mid, and High Risk categories.

Blood Sugar emerged as the strongest single marker, followed by systolic and diastolic blood pressure. The Decision Tree provided a substantially stronger baseline than Logistic Regression and achieved 0.836 High-Risk Recall and 0.920 High-Risk Precision. At the same time, the model struggled to distinguish Mid Risk from Low Risk and still missed 9 of 55 High-Risk test observations.

The main contribution of the project is therefore not a claim of clinical prediction readiness, but a transparent demonstration of how statistical reasoning can guide model development and interpretation. The results provide a useful baseline for future work with larger, clinically validated, and externally tested maternal-health datasets.

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